



## Research Article

## AI-based neural network models for bus passenger demand forecasting using smart card data

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## ABSTRACT

Accurate short-term forecasting of public transport demand is essential for the operation of on-demand public transport. Knowing where and when future demands for travel are expected allows operators to adjust timetables quickly, which helps improve service quality and reliability and attract more passengers to public transport. This study addresses this need by developing AI-based deep learning models for prediction of bus passenger demands based on actual patronage data obtained from the smart-card ticketing system in Melbourne. The models, which consider the temporal characteristics of travel demand for some of the heaviest bus routes in Melbourne, were developed using real-world data from 18 bus routes and 1,781 bus stops. LSTM and BiLSTM deep learning models were evaluated and compared with five conventional deep learning models using the same data set. A desktop comparison was also undertaken against a number of established demand forecasting models that have been reported in the literature over the past decade. The comparative evaluation results showed that BiLSTM models outperformed other models tested and was able to predict passenger demands with over 90% accuracy.

## 1. Introduction

A well-developed urban public transport system, especially bus transport, can reduce congestion and emissions and decrease the use of private vehicles (Li, Cao, et al., 2020). On-demand public transport, in particular, is seen to have the potential to improve operations further and enhance customer satisfaction. However, this type of service requires that short-term forecasts of future demands for bus services are known in advance (Liang et al., 2019; Liyanage & Dia, 2020; Liyanage et al., 2019; Smith et al., 2002; Zhou et al., 2013). Accurate prediction of future demands also helps operators to pre-allocate constrained resources such as vehicles and drivers to meet passenger demands and provide quality and reliable services with minimum waiting times. It also allows operators to optimize bus fleet management to minimise operational costs (Ma et al., 2014; Tirachini et al., 2013). Demand prediction is an integral part of business and commerce operations and helps decision makers to reduce the uncertainties of future operations. In public transport operations, the passenger service business models are highly dependent on accurate estimation of future passenger demands. Starting from route design and network planning, through to scheduling of vehicles with optimised seating capacity to meet operators' and users' objectives, and pricing each passenger vehicle on a network route, service operations in every planning horizon is dependent in one way or another on accurate estimation of future demands (Banerjee et al., 2020; Lu et al., 2021).

The focus of this paper is directly aligned with current advancements in digitalisation of urban mobility planning tasks. This is

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manifested through the paper's focus on development and evaluation of advanced passenger demand forecasting models which are important pillars in the planning and delivery of efficient and customer-focused on-demand public transport systems. First, the paper presents developments in app-based on-demand public transport services that are supported by technologies and advancements in IoT, Big Data and Real-Time Analytics. The paper then outlines how the success of these emerging modes of on-demand public transport, currently being run in trials in a number of cities around the world (including Sydney), relies heavily on accurate estimations of travel demand and passenger numbers over short durations up to 1 h. The paper then presents development and evaluation of advanced methodologies for passenger demand forecasting based on AI and deep learning theories that can be used for estimating future passenger demands. Another novel aspect of this work directly aligned to digitalisation of urban infrastructure is the use of smart card based field passenger demand data for model development. Unlike a large number of studies on this topic that used simulated data, this paper uses real-world field data obtained from IoT-enabled MyKi fare collection smart cards used for public transport in Melbourne.

Long-term public transport demand forecasting methodologies, including the well-established four-step models, elasticity and economic models, long-range demand models, mainly focus on planning issues and are not suited for operational forecasts (Balcombe, 2004; Ma et al., 2014). The research on short-term passenger demand forecasts for operational purposes is far more limited. Unlike the planning models, short-term passenger demand prediction models are used to account for short-term dynamic changes in demand due to weather conditions, traffic congestion, and special events. These models have gained widespread attention from transport planners and researchers in recent years due to their important operational role (Dia et al., 2001; Noekel, 2016; Zhai et al., 2018), making them a prerequisite for proactive operations and management of bus transport services (Ceder et al., 2013; Xue et al., 2015).

Demand prediction methodologies are based on understanding the temporal and spatial relationship between different variables in historical data. The primary objective is to obtain accurate and realistic forecasting of future demands (Zhai et al., 2020). Parametric and non-parametric methods are generally the two main solution techniques for short-term demand prediction techniques (Wei & Chen, 2012; Wu, Jiang, et al., 2020). In parametric approaches (also known as linear models or statistic approaches), the Box-Jenkins methodology is the most widely used model (Box et al., 2015). This model applies ARIMA or ARMA, AR, decision tree models to identify trends (Anvari et al., 2016; Cyril et al., 2018; Gan et al., 2014; Gong et al., 2014; Milenković et al., 2018; Wu, Xia, & Jin, 2020). However, these models are limited in scope because they are developed based on linear assumptions among time-lagged variables (Bai et al., 2017). Examples where these linear approaches were applied include two studies by (Ma et al., 2014) and (Xue et al., 2015) where the authors proposed Interactive Multiple Models that combine different algorithms to forecast passenger demands during different times of the day. In the first study (Ma et al., 2014), applied AR, SARIMA and ARIMA for weekly, daily, and hourly time-series analyses, respectively. In the second study (Xue et al., 2015), applied ARMA SARIMA and ARIMA models for weekly, daily and hourly time-series analyses, respectively. However, the linear models depend on high-quality data comprising accurate and non-fluctuating patterns to develop a time-series sequence. However, real-world passenger demand data are random and unstable, which renders linear approaches incapable of describing the variations in passenger flows. Hence, other methodologies have been presented in the literature to track such non-linear characteristics.

The second category, non-linear or non-parametric approaches, constructs non-linear relationships between input and output variables (Wu, Li, et al., 2020). There are also hybrid approaches that combine multiple algorithms strategically. Non-parametric models include support vector machine (SVM) (Chen et al., 2015; Jiang et al., 2014; Yang & Liu, 2016), least-squares SVM (Guo et al., 2013; Sun et al., 2015; Zhang et al., 2011), Fuzzy neural networks (Buckley & Hayashi, 1994; Dou et al., 2013; Tsai et al., 2009), Bayesian networks (Roos et al., 2017; Sun et al., 2006), grey models (Hai-lan & En-chong, 2012; Wang & Zhang, 2012; Yang et al., 2013; Zhang et al., 2017) and neural networks (Chen et al., 2012; Pekel & Soner Kara, 2017; Zhang et al., 2013). Among these, Artificial Intelligence (AI) approaches, based on deep learning neural network methodologies, have been identified as most promising and practical for complex time series forecasts (Lee et al., 2006) and have been shown to provide improved predictive capability (Vlahogianni & Karlaftis, 2011).

Other recent literature on passenger demand forecasting also examined the relationship between Internet of Things sensing infrastructures, intelligent vehicular networks, and big data-driven algorithmic decision-making for provision of accurate short-term forecasting of public transport demand (Aldridge & Stehel, 2021; Blake & Michalikova, 2021; Lăzăroi & Harrison, 2021; Nica, 2021). Recent literature on passenger demand forecasting also examined the relationship between deep learning-based sensing technologies, predictive control algorithms, and big geospatial data analytics (Adams et al., 2021; Pelau et al., 2021; Rowland & Porter, 2021; Wallace & Lăzăroi, 2021). Along with the above, AI-based neural networks have attracted research interest in a large number of transport application areas (Dia, 2001; Dia et al., 1996; Dia & Panwai, 2011; Dia & Rose, 1995; Thomas & Dia, 2006; Thomas et al., 2001). Recently, deep learning neural network innovations have also attracted research interest in the context of demand prediction. For example, in a study by (Liu et al., 2019, 2020), the authors predicted short-term metro passenger flows using end-to-end deep learning architectures using multiple sources of data. A “multi-pattern deep fusion (MPDF)” approach was proposed by (Bai et al., 2017), which was constructed through fusing “Deep Belief Networks (DBNs)” to represent passenger flow, where the DBNs were proven to be successful in extracting complex features (Li, Cabrera, et al., 2016). An “End-to-end Multi-task Learning Temporal CNN” was also proposed by (Zhang et al., 2019) to forecast short-term passenger demand at a multi-zone levels. A hybrid estimation model was introduced based on CNN and spatial-temporal contexts (Liu & Chen, 2017; Liu et al., 2017). In their model, the CNN detected passengers, while the spatial-temporal model accurately predicted passenger volumes. Another study captured spatial-temporal correlations of demands using a “Fusion Convolutional-LSTM deep learning” approach (Ke et al., 2017). Also (Han et al., 2019), proposed an optimised hybrid LSTM model for accurate bus passenger flow prediction, which was shown to have excellent performance accuracies. LSTM models have also been used in spatial-temporal aspects to forecast dynamic origin-destination matrices in a subway network in France (Toqué et al., 2016). In another study, a spatial-temporal model was redesigned for rail transit passenger demand estimation based on multiple data sources such as smart card, weather and mobile phone data (Li, Cao, et al., 2020). Similarly, origin-destination matrices prediction was tested in the context of ride-hailing or taxi passenger demand utilising various non-parametric models including LSTM approaches

**Table 1**  
Description of the fields of the dataset.

| Field                | Description                                                                                                                                                                                                                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Smart card touch-on  | Transaction time of touch-on, including date, in POSIX format (AEST)                                                                                                                                                                     |
| Smart card touch-off | Transaction time of touch-off, including date, in POSIX format (AEST)                                                                                                                                                                    |
| Origin bus stop      | Stop ID of boarding                                                                                                                                                                                                                      |
| Route number         | Route ID of boarding                                                                                                                                                                                                                     |
| Route direction      | Route direction of boarding                                                                                                                                                                                                              |
| Destination bus stop | Stop ID of alighting                                                                                                                                                                                                                     |
| Paired type          | Not every touch-on has a corresponding touch-off. Some touch-offs are estimated through a model that uses existing 'paired' trips to infer the likely 'synthetic' touch-off. Trips that are 'paired' are more reliable than 'synthetic.' |
| Boost factor         | Boost factor used to scale up the transaction to approximate actual patronage                                                                                                                                                            |

(Chen et al., 2020; He et al., 2022; Ke et al., 2021; Wang et al., 2019; Zhang, Che, et al., 2021; Zhang, Che, et al., 2021).

Based on the results reported in previous studies, most parametric and non-parametric methods faced major challenges in producing the high accuracies (90% and above) required for reliable real-world operations. This study bridges these gaps by developing deep learning passenger bus transport demand forecasting models with required accuracy in the short-term context (15-min, 30-min and 60-min time horizons). As shown in the results section, the accuracy of our models exceeded 90% for all tested scenarios. Generally, neural network models' performance is challenged by difficulties in feature learning and complexities in pattern recognition (Panwai & Dia, 2006). Researchers have used data pre-processing for pattern recognition (Bai et al., 2017; Panwai & Dia, 2005; Rose & Dia, 1995) and intensive networks for feature learning (Hinton et al., 2006). In our study, which utilised the neural network approach to predict passenger demand on bus routes, time-series decomposition is used to pre-process the data and to understand various patterns. Deep learning neural networks (i.e. networks with more than one hidden layer) were then used for intensive feature learning (Bengio, 2009; Bengio et al., 2007; Li, Bai, & Zeng, 2016).

From a practical perspective, and in addition to the academic contributions of this paper, it is important to note that passenger demand forecasting is an integral part of real-world public transport operations. Decision makers need to have accurate estimates of what the demand for travel will be particularly for short-term horizons. Knowing these patterns of demands helps operators to optimize operations, reduce costs and enhance customer satisfaction. This study is therefore directly related to addressing this industry challenge by using state-of-the-art AI-based deep learning forecasting models that improve passenger bus transport demand forecasting for short-term horizons up to 60 min into the future. The model developments and enhancements discussed in this study, which represent meaningful advancements on today's best practice algorithms and solutions, have both a technical and academic significance in addition to important practical significance to the industry.

This study aims to improve passenger bus transport demand forecasting in the short-term context (15-min, 30-min and 60-min time horizons). The key research gap addressed in this paper is the lack of accurate public transport passenger demand forecasting models that can be used reliably for real-world applications. While previous literature relied on development of statistical-based forecasting models using simulation data, this paper presents state-of-the-art deep learning models that were developed using real-world IoT-enabled smart card fare collection data reflecting real-world conditions. This contributed to development of highly reliable and accurate models compared to methodologies described in existing literature.

The remainder of this paper includes the following sections. The second section presents the data sources used for model development, specification of the study area and data requirements, methodological framework, time-series pattern development and necessary data preparations to facilitate demand prediction. Section 3 presents the LSTM model development for selected bus routes and reports on a comparative analysis of a number of neural network models. The fourth section presents an analysis of results and compares the performance of the tested models. Finally, the fifth section presents conclusions and insights for further research work.

## 2. Data source and methodology

This study relied on passenger demand data from Melbourne's MyKi smart card ticketing system, which is a reloadable contactless smart card launched in 2008. The system became fully operational for electronic payment of public transport fares in 2012. The Myki card system is convenient for passengers with easy access to fare payment devices that are installed in public service vehicles. Passengers can immediately pay for their trips by touching on and off from their start and end locations. For this study, the authors' accessed one-month data, May 2018, from the Department of Transport in Victoria.

### 2.1. Data format and fields

The MyKi data is collected from recorded passenger touch-on and touch-off events and used to obtain passenger boarding and alighting counts for bus service routes. This data and information about Metropolitan Melbourne bus routes in the study area were also obtained. The dataset received for this study contained the fields presented in Table 1. Approximately 10–14% of bus transactions occurred on transport vehicles where MyKi was operating in a "headless mode", meaning that the service operator was known, but the

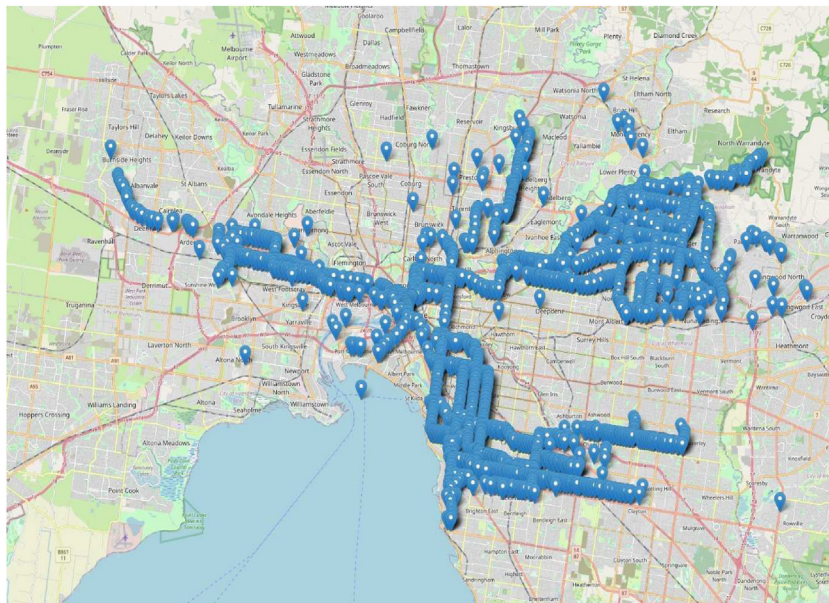


Fig. 1. Distribution of bus stop locations covering 18 routes in the study area.

Table 2

Details of the bus routes operating in the study area.

| Route ID | Service Direction                                                 | Number of bus stops | Total trips |
|----------|-------------------------------------------------------------------|---------------------|-------------|
| 200      | City (Queens Street) via Kew Junction                             | 58                  | 44,237      |
|          | Bulleen via Kew Junction                                          | 60                  |             |
| 207      | City (Queens Street) via Kew Junction                             | 58                  | 50,445      |
|          | Doncaster SC via Kew Junction                                     | 61                  |             |
| 216      | City (Queens Street) Brighton Beach                               | 87                  | 18,932      |
|          | Sunshine Station                                                  | 86                  |             |
| 219      | Gardenvale via City                                               | 87                  | 19,496      |
|          | Sunshine South via City                                           | 92                  |             |
| 220      | Gardenvale via City                                               | 88                  | 42,868      |
|          | Sunshine via City                                                 | 88                  |             |
| 246      | Clifton Hill via St Kilda                                         | 45                  | 65,541      |
|          | Elsternwick via St Kilda                                          | 44                  |             |
| 302      | City (Lonsdale St) via Belmore Rd > Eastern Freeway               | 51                  | 42,917      |
|          | Box Hill via Eastern Freeway > Belmore Road                       | 47                  |             |
| 304      | City (Lonsdale St) via Belmore Rd > Eastern Freeway               | 47                  | 26,001      |
|          | Doncaster SC via Eastern Freeway > Belmore Road                   | 45                  |             |
| 305      | City (King/Lonsdale Streets) via Eastern Freeway                  | 50                  | 24,132      |
|          | The Pines via Eastern Freeway                                     | 48                  |             |
| 309      | Donvale via Eastern Freeway > Reynolds Road                       | 53                  | 10,395      |
|          | City (Queens street) via Reynolds Road > Eastern Freeway          | 54                  |             |
| 350      | La Trobe University via Eastern Freeway                           | 41                  | 12,679      |
|          | City (Queen Street) via Eastern Freeway                           | 45                  |             |
| 605      | Flagstaff Station via Kooyong Road                                | 53                  | 13,337      |
|          | Gardenvale via Kooyong Road                                       | 49                  |             |
| 623      | St Kilda via Chadstone SC                                         | 72                  | 28,674      |
|          | Glen Waverley via Chadstone SC                                    | 73                  |             |
| 625      | Elsternwick via Oakleigh > Ormond                                 | 50                  | 9,652       |
|          | Chadstone SC via Ormond > Oakleigh                                | 51                  |             |
| 630      | Elwood via Ormond                                                 | 54                  | 57,292      |
|          | Monash University via Ormond                                      | 53                  |             |
| 905      | City (King/Lonsdale Streets) via Templestowe > Eastern Freeway    | 47                  | 56,409      |
|          | The Pines via Eastern Freeway > Templestowe                       | 43                  |             |
| 906      | City (Kind/Lonsdale Streets) via The Pines SC                     | 51                  | 68,906      |
|          | Warrandyte via The Pines SC                                       | 48                  |             |
| 907      | City (Kind/Lonsdale Streets) via Doncaster Road > Eastern Freeway | 47                  | 101,776     |
|          | Mitcham via Eastern Freeway > Doncaster Road                      | 46                  |             |

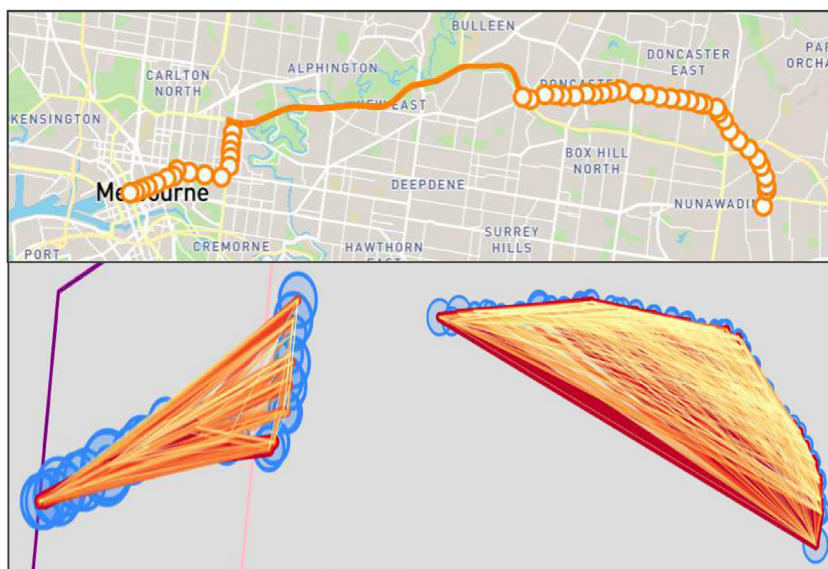


Fig. 2. (a) Map of movement of route 907 in both service directions (b) Bus trip distribution among bus stops along route 907.

route was unknown. It is also estimated that approximately 60% of passengers generally touch their cards when using buses, but a lower percentage touches off. As a result, while smart card readers may not record every traveller's information, the data can still be used to reflect and understand passenger travel demand. The raw passenger origin-destination data was supplied to the research team in an anonymised format. School buses were excluded from this analysis.

To account for passengers who may not touch on or off, missing passenger patronage was estimated by the Department of Transport using surveys to supplement smart card data. Passengers may not touch on for various reasons, such as faulty smart cards, faulty smart card readers or fare evasion. The survey data was used to arrive at a weighting (or boost factor) associated with each transaction to approximate actual bus patronage. The dataset was thoroughly checked, and all invalid trips were removed.

The smart card data provided a good understanding of existing bus passenger demand profiles and patterns, which in turn was sufficient for producing accurate and reliable prediction results. The data set included information for the entire bus route (18-service routes with both service directions for each route) and 1,781 touch-on bus stops. The specific routes included routes 200, 207, 246, 302, 304, 305, 309, 350, 905, 906, 907, 605, 623, 630, 220, 216, 219, and 625 in both service directions. The operating timetables for existing bus services in Metropolitan Melbourne (during which the MyKi data was obtained) were extracted from Public Transport Victoria travel information portal. They included separate schedules for weekdays, Saturdays and Sundays (PTV, 2019).

## 2.2. Study area and data description

The city of Melbourne in Australia was selected as the case study for this research. Melbourne has one of Australia's largest urban population, comprising more than 5 million people (World population review, 2021). Each blue colour place marker in Fig. 1 represents an existing bus stop for the 18 routes covered in the available data across the Southeast region of Melbourne.

The study area included a number of inner suburbs in the southeast of Melbourne, which covered significant bottleneck arterials that are among the most congested corridors in Melbourne. The case study area is also well-served by different types of public transport, including fixed-service buses equipped with MyKi smart-card readers. Among 715,375 trips, 693,689 trips were useable (96.97%) for this analysis. The study area covered 18-service routes and 1,781 bus stops in the Metropolitan Melbourne region, summarised in Table 2.

In the Metropolitan Melbourne area, bus services in both service directions move intensively between 5:00 a.m. to 1:00 a.m. The traffic maps for each of the 18 bus routes were developed to visualise where they run within the study area and to also understand the total passenger trip distributions between bus stops. For example, Route 907 (the most attractive public bus transport movement among the 18 service routes) is selected to present a zoomed view of the details (Fig. 2). Route 907 is shown in Fig. 2a, where each stop is marked with a circle whose size is proportional to number of trips originating from and ending at that stop. The width of the line linking each origin to destination represents the number of passenger journeys between the bus stops. There were 78 service vehicles on the 907-route and 77 vehicles on the 907b route for an average weekday. Route 907 is 36 km in length and traverses mainly arterials and the Eastern Freeway. Residential communities, schools and workplaces are located along this bus route, accommodating many work and education trips on weekdays.

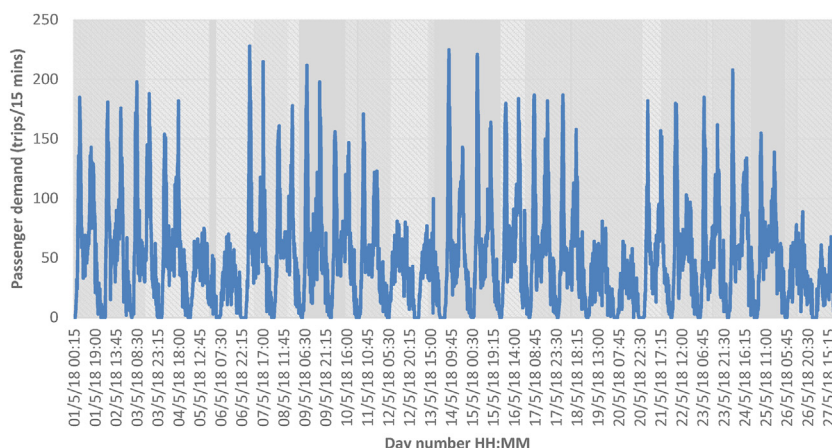


Fig. 3. Passenger demand for Route 907 in May 2018.

Table 3

Heat map of passenger counts for route 907.

| Date/Time | 1:00 | 6:00 | 7:00 | 8:00 | 9:00 | 10:00 | 11:00 | 12:00 | 13:00 | 14:00 | 15:00 | 16:00 | 17:00 | 18:00 | 19:00 | 20:00 | 21:00 | 22:00 | 23:00 | 0:00 |
|-----------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
| Monday    | 0    | 19   | 113  | 390  | 436  | 191   | 156   | 108   | 131   | 140   | 167   | 278   | 276   | 404   | 240   | 125   | 98    | 69    | 37    | 23   |
| Tuesday   | 2    | 33   | 162  | 514  | 613  | 257   | 206   | 190   | 225   | 223   | 249   | 350   | 364   | 465   | 299   | 168   | 113   | 79    | 58    | 23   |
| Wednesday | 7    | 27   | 160  | 588  | 528  | 191   | 194   | 180   | 218   | 242   | 243   | 347   | 387   | 566   | 317   | 136   | 133   | 132   | 83    | 52   |
| Thursday  | 8    | 26   | 154  | 538  | 537  | 201   | 222   | 194   | 215   | 225   | 254   | 422   | 368   | 520   | 285   | 132   | 136   | 118   | 55    | 24   |
| Friday    | 11   | 23   | 140  | 468  | 500  | 222   | 224   | 177   | 200   | 225   | 225   | 376   | 360   | 463   | 276   | 146   | 150   | 157   | 86    | 79   |
| Saturday  | 26   | 0    | 36   | 85   | 161  | 165   | 193   | 206   | 216   | 189   | 183   | 192   | 198   | 224   | 158   | 107   | 75    | 69    | 75    | 36   |
| Sunday    | 5    | 0    | 2    | 49   | 113  | 139   | 189   | 204   | 160   | 139   | 137   | 143   | 161   | 172   | 88    | 76    | 83    | 21    | 0     | 0    |

### 2.3. Data structure and methodology

The methodology adopted in this study is based on an empirical data-driven framework for predicting bus passenger demands using neural networks. The first step includes dividing the full data set into three-time series, namely 15, 30 and 60-min. The second step was to develop corresponding neural network models to capture the series features and generate reliable and accurate prediction outputs. The demand represents the number of passengers in both directions.

The historical data are aggregated in 15 min, as 15 min is the most extended departure interval experienced for all 18 routes during off-peak periods. There were 80-time intervals between 05:00 to 24:00 + 1 (i.e. 1:00 the following day). To better understand the characteristics of passenger demand, the historical data in May 2018 was plotted for all 18 routes.

Fig. 3 provides an example for Route 907. There are no public or school holidays in this selected month of 2018. The line diagram in Fig. 2 shows passenger demand fluctuation represents cyclical variations that follow a similar pattern almost every week for all weekdays and weekends. It is interesting to note that significant gaps in passenger demand are observed from peak to off-peak periods on weekdays, and from weekdays to weekends due to reduced work and study trips. Furthermore, pronounced asymmetries of demand of bus passengers were identified between morning and evening peaks. Among these, the morning peak is characterised by sharp and pronounced spikes compared to the evening peak because work and study journeys usually coincide in the morning. In contrast, the afternoon peak is flatter and more spread due to earlier return-to-home study trips. Not surprisingly, demand fluctuates less during weekends.

Table 3 shows a heat map of the number of passengers per hour averaged over seven days of the week for route 907. The colours indicate where the value sits in different ranges: Red represents the lowest passenger count during each hour of an average day, while green depicts the highest passenger count.

Three prediction models are developed for 15, 30, and 60 min prediction horizons. A description of the adopted for this work is presented next.

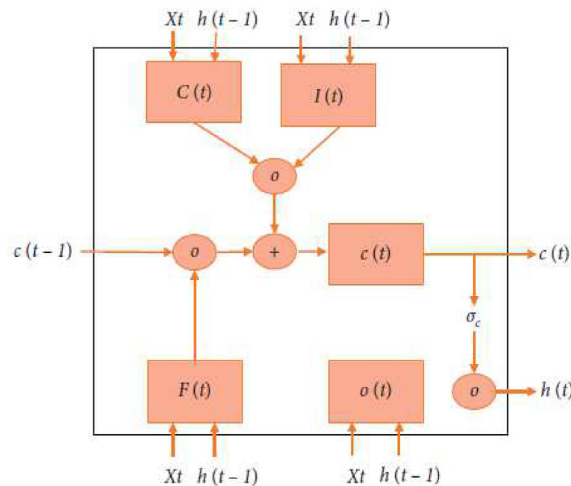
A vector of a single row of elements for the time series for day  $n$  is represented as  $d^n = \{d^n(t) | t = 1, 2, \dots, N_{Tin}\}$ . In this,  $d^n(t)$  is the bus passenger demand at a time interval  $(t - 1, t]$  for day  $n$ ,  $t \in [1, 80]$ . Here,  $t$  represents an integer and  $N_{Tin}$  the number of time intervals in a day (80-time intervals for this study).

Then in the source dataset, a time series is defined as  $d_s = [d^1, d^2, \dots, d^{N_{day}}]$ .  $N_{day}$  represents the full days covered in the dataset, ranging from 1st to 28th May for this study. Relevant pattern time series of three different time series was constructed, which are 15-

**Table 4**

Demand datasets used for each prediction horizon.

| Time horizon | Total dataset | Training dataset | Test and validation dataset |
|--------------|---------------|------------------|-----------------------------|
| 15-min       | 27,823        | 16,694           | 11,129                      |
| 30-min       | 27,360        | 16,416           | 10,944                      |
| 60-min       | 26,561        | 15,937           | 10,624                      |

**Fig. 4.** LSTM model architecture.

min, 30-min, and hourly.

1. For 15min:  $TS_{15m}^n(t)$  consists of bus demand data  $d^n(t)$  in  $n_{15m}$  time intervals.

$$TS_{15m}^n(t) = \{d^n(t-1), d^n(t-2), \dots, d^n(t-n_{15m})\}$$

2. For 30min:  $TS_{30m}^n(t)$  consists of bus demand data  $d^n(t)$  in  $n_{30m}$  time intervals.

$$TS_{30m}^n(t) = \{d^n(t-1), d^n(t-2), \dots, d^n(t-n_{30m})\}$$

3. For 60min:  $TS_h^n(t)$  consists of bus demand data  $d^n(t)$  in  $n_h$  time intervals.

$$TS_h^n(t) = \{d^n(t-1), d^n(t-2), \dots, d^n(t-n_h)\}$$

#### 2.4. Data preparation for neural network modelling

Passenger demand data collected from the smart card data system were aggregated for different durations such as 15 min, 30 min, and 1 h. It is important to note that temporal data aggregation can alter the underlying characteristics and features of the source field data (Vlahogianni & Karlaftis, 2011). For this study, 15-min data aggregation was selected because it was the maximum interval for bus departures during off-peak periods. The 15 min aggregated passenger demand was organised by day and by week from Monday to Sunday over the entire month. Among the 80-time intervals, from 05:00 to 24:00 + 1 (i.e. 1:00 the following day), erroneous samples and time intervals with significantly less passenger demand were excluded. Therefore, only time intervals from morning 07:00 to evening 21:30 within a day were considered. Model development included dividing the available data into “training and testing” data sets. The training set was approximately 70% of the whole dataset and was used to develop the model to calibrate the model parameters. The testing data set comprised about 30% of the total dataset used for the model validation and to test the demand prediction accuracy of the developed models. The data sets used in every three models are presented in Table 4.

Neural network models work poorly with data sets that have high variability. For example, in our dataset, passenger demand changes between 0 to around 250 passengers' trips per 15-min. Such heterogeneity of data can complicate the learning process (Faridai et al., 2021). The heterogeneity of data was addressed using data smoothing, which was used to detect trends in noisy data in cases where the shape of the movement is unknown. Exponential smoothing was used with a 0.5 damping factor.

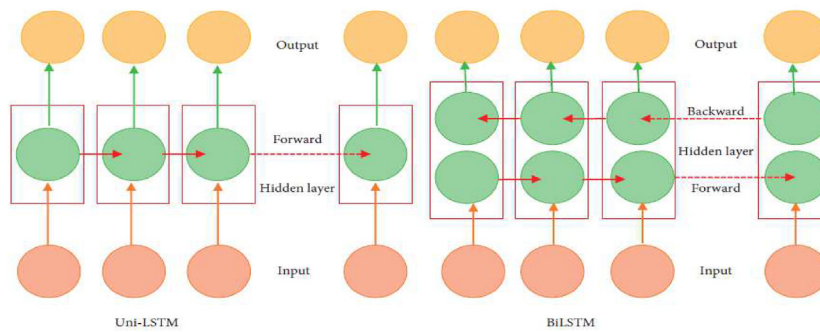


Fig. 5. Uni-LSTM and BiLSTM architectures.

Table 5  
LSTM model parameters.

| Parameters                             | Settings                             |
|----------------------------------------|--------------------------------------|
| Gradient Decay Factor                  | 0.9                                  |
| Initial Learning Rate                  | $5 \times 10^{-3}$                   |
| Minimum Batch Size                     | 128                                  |
| Maximum Epochs                         | 300                                  |
| Training Optimizer                     | Adaptive moment estimation optimizer |
| Dropping Learning Rate During Training | Piecewise                            |
| Learning Rate Drop Period              | 125                                  |
| The factor for Learning Rate Dropping  | 0.2                                  |

### 3. Artificial neural network model development

Recurrent Neural Networks (RNNs) architectures were in the past found to provide good prediction accuracy. However, with the inability to use information from a distant past, RNNs did not perform well for long-term memory (Abduljabbar, Dia, Tsai, et al., 2021). The LSTM models address this problem as they are considered extensions of the RNNs (Hochreiter & Schmidhuber, 1997). LSTM models have the capability of learning patterns with long dependencies compared to traditional RNNs that are not known for performing well for longer horizon patterns. Hence, LSTM models are generally identified to outperform RNNs in time series data forecasting (Yeon et al., 2019). Uni-directional and bi-directional LSTM models were developed in this study. Bi-directional LSTM (BiLSTM) is an extension to the unidirectional LSTM that includes additional data training where the model trains in forward and backward directions (Abduljabbar, Dia, & Tsai, 2021). The model architectures are presented below. LSTM models have different structures than RNN models, as presented in Fig. 4 and Fig. 5. The following formulae calculate the predicted values.

$$\text{Input gate } (I_t) = \sigma_g(W_i X_t + R_i h_{t-1} + b_i) \quad (1)$$

$$\text{Forget gate } (f_t) = \sigma_g(W_f X_t + R_f h_{t-1} + b_f) \quad (2)$$

$$\text{Cell Candidate } (C_t) = \sigma_g(W_c X_t + R_c h_{t-1} + b_c) \quad (3)$$

$$\text{Output gate } (O_t) = \sigma_g(W_o X_t + R_o h_{t-1} + b_o) \quad (4)$$

Where:

$\sigma_g$  - Gate activation function

$W_{i,f,c,o}$  - Input weight matrices

$R_{i,f,c,o}$  - Recurrent weight matrices

$X_t$  - Input

$h_{t-1}$  - Output at the previous time  $t - 1$

$b_{i,f,c,o}$  - Bias vectors

The forget gate determines the “levels of prior memory that need to be removed from the cell state” (Yeon et al., 2019). In similar manner, the input gate specifies new (cell state) inputs. Then, the cell state  $C_t$  and the output  $H_t$  at time,  $t$  is calculated as follows for the LSTM.

**Table 6**

Neural network architectures and parameters.

| Model Architecture                       | Model Parameters               | Learning Rule          | Transfer Function |
|------------------------------------------|--------------------------------|------------------------|-------------------|
| Deep Learning Backpropagation (DLBP)     | LCoef (Learning rate)          | Delta-Rule             | Linear            |
|                                          | Hidden layer 1–0.3             | Norm-Cum-Delta         | <b>TanH</b>       |
|                                          | Hidden layer 2–0.25            | Ext DBD                | Sigmoid           |
|                                          | Hidden layer 3–0.2             | Quick Prop             | DNNA              |
|                                          | Output – 0.15                  | Max-Prop               | Sine              |
|                                          | Momentum 0.4                   | Delta-Bar-Delta        |                   |
|                                          | Trans. Pt. 10000               |                        |                   |
|                                          | LCoef Ratio 0.5                |                        |                   |
|                                          | F' Offset 0.1                  |                        |                   |
|                                          | Epoch 16                       |                        |                   |
| Modular Neural Network (MNN)             | LCoef (Learning rate)          | Delta-Rule             | Linear            |
|                                          | Hidden layer 1–0.3             | Norm-Cum-Delta         | <b>TanH</b>       |
|                                          | Output – 0.15                  | Ext DBD                | Sigmoid           |
|                                          | Momentum 0.4                   | Quick Prop             | DNNA              |
|                                          | Trans. Pt. 10000               | Max-Prop               | Sine              |
|                                          | LCoef Ratio 0.5                | <b>Delta-Bar-Delta</b> |                   |
|                                          | F' Offset 0.1                  |                        |                   |
|                                          | Epoch 4                        |                        |                   |
| Radial Basis Function Network (RBFN)     | LCoef (Learning rate)          | Delta-Rule             | Linear            |
|                                          | Proto – 0.3                    | Norm-Cum-Delta         | <b>TanH</b>       |
|                                          | Hidden layer 2–0.25            | Ext DBD                | Sigmoid           |
|                                          | Output – 0.15                  | Quick Prop             | DNNA              |
|                                          | Momentum 0.4                   | Max-Prop               | Sine              |
|                                          | Trans. Pt. 10000               | Delta-Bar-Delta        |                   |
|                                          | LCoef Ratio 0.5                |                        |                   |
|                                          | Map Trans. 2000                |                        |                   |
|                                          | Epoch 16                       |                        |                   |
|                                          | Summation Function - Euclidean |                        |                   |
| General Regression Neural Network (GRNN) | Tau (time constant) – 1000     | NA                     | NA                |
|                                          | Reset Factor – 0               |                        |                   |
|                                          | Radius of Influence – 0.25     |                        |                   |
|                                          | Sigma Scale – 1                |                        |                   |
|                                          | Sigma Exponent – 0.5           |                        |                   |
|                                          | Summation Function – Euclidean |                        |                   |
|                                          | Epoch 16                       |                        |                   |
| Recurrent Neural Networks (RNNs)         | LCoef (Learning rate)          | Delta-Rule             | Linear            |
|                                          | Hidden layer 1–0.3             | Norm-Cum-Delta         | TanH              |
|                                          | Hidden layer 2–0.25            | Ext DBD                | <b>Sigmoid</b>    |
|                                          | Hidden layer 3–0.2             | Quick Prop             | DNNA              |
|                                          | Output – 0.15                  | Max-Prop               | Sine              |
|                                          | Momentum 0.4                   | Delta-Bar-Delta        |                   |
|                                          | Trans. Pt. 10000               |                        |                   |
|                                          | LCoef Ratio 0.5                |                        |                   |
|                                          | F' Offset 0.1                  |                        |                   |
|                                          | Epoch 16                       |                        |                   |

$$C_t = ft \odot ct \odot 1 + it \odot gt \quad (5)$$

$$H_t = ot \odot oc(ct) \quad (6)$$

Where;  $\odot$  represents the Hadamard product. This product denotes the element-wise vector multiplication.

For this work, the uni-directional and bi-directional LSTM models were developed in Matlab (Matlab, 2021). The data for short-term temporal prediction of bus routes were first arranged as two-column values: passenger demand at the time (t) and expected passenger demand (t + n), where n was 15, 30 and 60-min. The training (70% of the whole dataset) and testing (30% of the whole dataset) were standardised to prevent the model from overfitting. The developed LSTM network consisted of four layers, including a sequence input layer with one feature, uni-LSTM and bi-LSTM layers with 300 hidden units, a fully connected layer with one response and a regression layer. This developed model can be used in future research work for short-term demand prediction for any aspect where the dataset shows a time-series pattern.

The model settings are as presented in Table 5. Several trials were undertaken to obtain the most-suited combination of values to reach the highest accuracy. For the state activation function, tanH is used, and the sigmoid function is used for the gate activation function. The uniLSTM and biLSTM model development experiments were developed with Deep Learning Toolbox in Matlab R2019b.

### 3.1. Model evaluation

In order to evaluate the robustness of the LSTM prediction model, the accuracy of the prediction is compared with five other neural

**Table 7**  
Short-term demand prediction accuracies for bus routes.

| Bus Route                                     |        |          | 15-minutes |        |        | 30-minutes |        |        | Hourly |        |        |
|-----------------------------------------------|--------|----------|------------|--------|--------|------------|--------|--------|--------|--------|--------|
| Prediction horizon                            |        |          | MAPE       | MAE    | RMSE   | MAPE       | MAE    | RMSE   | MAPE   | MAE    | RMSE   |
| Temporal prediction accuracies for bus routes | BiLSTM | Error    | 2.99%      | 0.78%  | 2.57%  | 5.23%      | 1.60%  | 3.81%  | 12.39% | 5.62%  | 8.19%  |
|                                               |        | Accuracy | 97.01%     | 99.22% | 97.43% | 94.77%     | 98.40% | 96.19% | 87.61% | 94.38% | 91.81% |
|                                               | LSTM   | Error    | 22.51%     | 14.32% | 20.11% | 30.02%     | 17.85% | 24.98% | 35.28% | 18.74% | 25.71% |
|                                               |        | Accuracy | 77.49%     | 85.68% | 79.89% | 69.98%     | 82.15% | 75.02% | 64.72% | 81.26% | 74.29% |
|                                               | RNN    | Error    | 25.92%     | 19.63% | 29.88% | 34.91%     | 28.65% | 46.34% | 48.66% | 42.88% | 72.45% |
|                                               |        | Accuracy | 74.08%     | 80.37% | 70.12% | 65.09%     | 71.35% | 53.66% | 51.34% | 57.12% | 27.55% |
|                                               | DLBP   | Error    | 23.79%     | 19.71% | 30.20% | 54.21%     | 32.90% | 48.23% | 45.06% | 42.47% | 71.94% |
|                                               |        | Accuracy | 76.21%     | 80.29% | 69.80% | 45.79%     | 67.10% | 51.77% | 54.94% | 57.53% | 28.06% |
|                                               | MNN    | Error    | 23.79%     | 19.71% | 30.20% | 54.21%     | 32.90% | 48.23% | 45.06% | 42.47% | 71.94% |
|                                               |        | Accuracy | 72.76%     | 80.04% | 69.22% | 67.84%     | 70.48% | 51.35% | 51.88% | 59.33% | 31.87% |
|                                               | RBF    | Error    | 26.73%     | 23.77% | 46.13% | 32.90%     | 32.59% | 56.52% | 59.53% | 46.63% | 77.04% |
|                                               |        | Accuracy | 73.27%     | 76.23% | 53.87% | 67.10%     | 67.41% | 43.48% | 40.47% | 53.37% | 22.96% |
|                                               | GRNN   | Error    | 71.01%     | 30.50% | 49.51% | 78.33%     | 38.04% | 58.96% | 72.40% | 46.31% | 75.17% |
|                                               |        | Accuracy | 28.99%     | 69.50% | 50.49% | 21.67%     | 61.96% | 41.04% | 27.60% | 53.69% | 24.83% |

network architectures based on the same data set. These included “Deep Learning Back-Propagation (DLBP)”, “Modular Neural Networks (MNNs)”, “Radial Basis Function Networks (RBFNs)”, “General Regression Neural Networks (GRNNs)” and “Recurrent Neural Networks (RNNs)”. These models are widely used in literature to forecast different attributes in the transport network. The five models used for comparative analysis were developed using the NeuralWorks Professional II software, a commercial package for neural network model development systems (NeuralWare, 2021). These models are well-established in the literature and the reader is referred to (Jacobs et al., 1991) for more details.

The parameters selected in this study and the learning rules and transfer function combinations that resulted in best performance are presented in Table 6.

### 3.2. Performance measures

Finding multiple suitable indicators for measuring prediction accuracy is essential, as no single indicator can measure the full precision/bias of predicted values. The following performance indicators were used to compare models (Vandeput, 2019). Therefore, for this analysis prediction accuracy is compared in three performance measures; MAPE, MAE and RMSE.

#### 1. Mean Absolute Percentage Error (MAPE)

MAPE is a common key performance indicator used for prediction accuracy. MAPE divides each error by respective demand as in equation (7). In this measure, high errors during low-demand periods can significantly impact MAPE.

$$MAPE = \frac{1}{n} \sum \frac{|predicted\ value - demand|}{demand} \quad (7)$$

#### 2. Mean Absolute Error (MAE) %

MAE is a good performance indicator also used to measure prediction accuracy. Key limitation is that it is not scaled to average demand. Therefore, MAE is divided by average passenger demand to obtain MAE % as in equation (8).

$$MAE\ \% = \frac{\sum |predicted\ value - demand|}{\sum demand} \quad (8)$$

#### 3. Root Mean Squared Error (RMSE)

RMSE is a valuable key indicator calculated using Equation (9). RMSE is not scaled to the demand. Therefore, it is divided by average

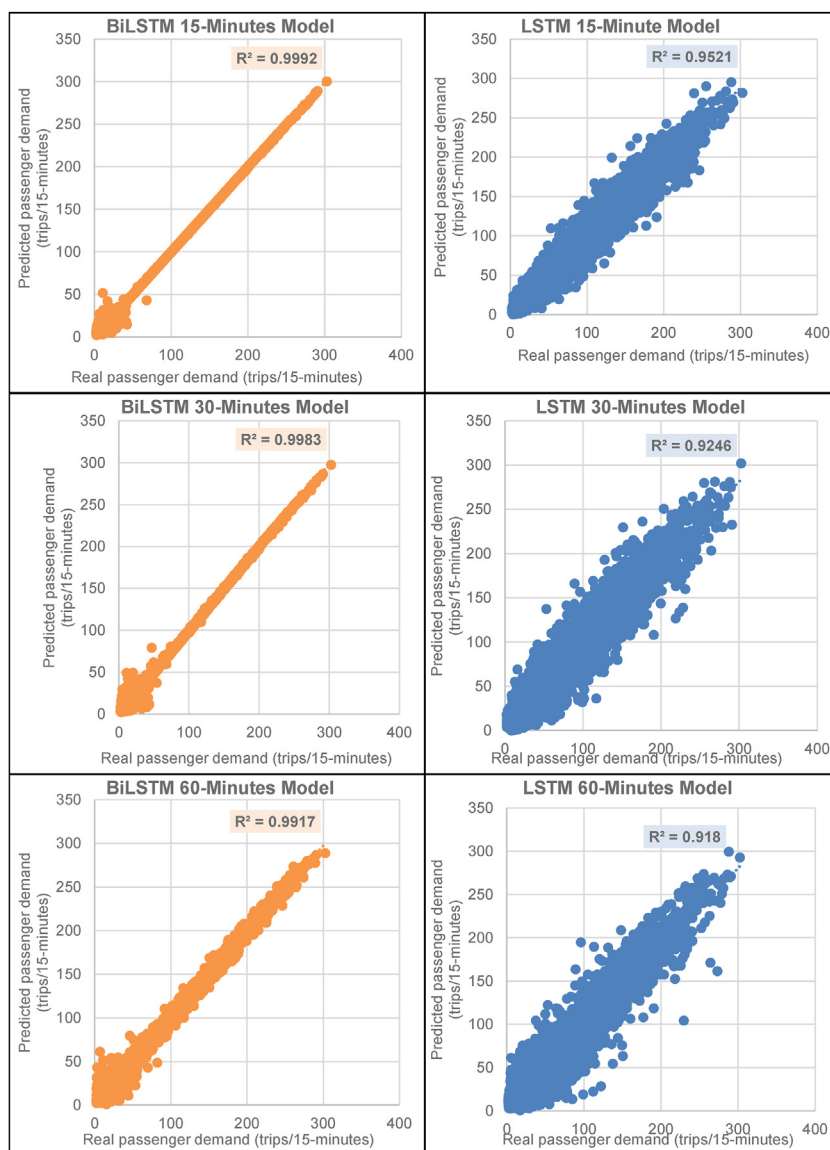


Fig. 6. Correlation between actual passenger demand to predicted passenger demand for BiLSTM and LSTM models.

demand to obtain RMSE %.

$$RMSE = \sqrt{\frac{1}{n} \sum (predicted\ value - demand)^2} \quad (9)$$

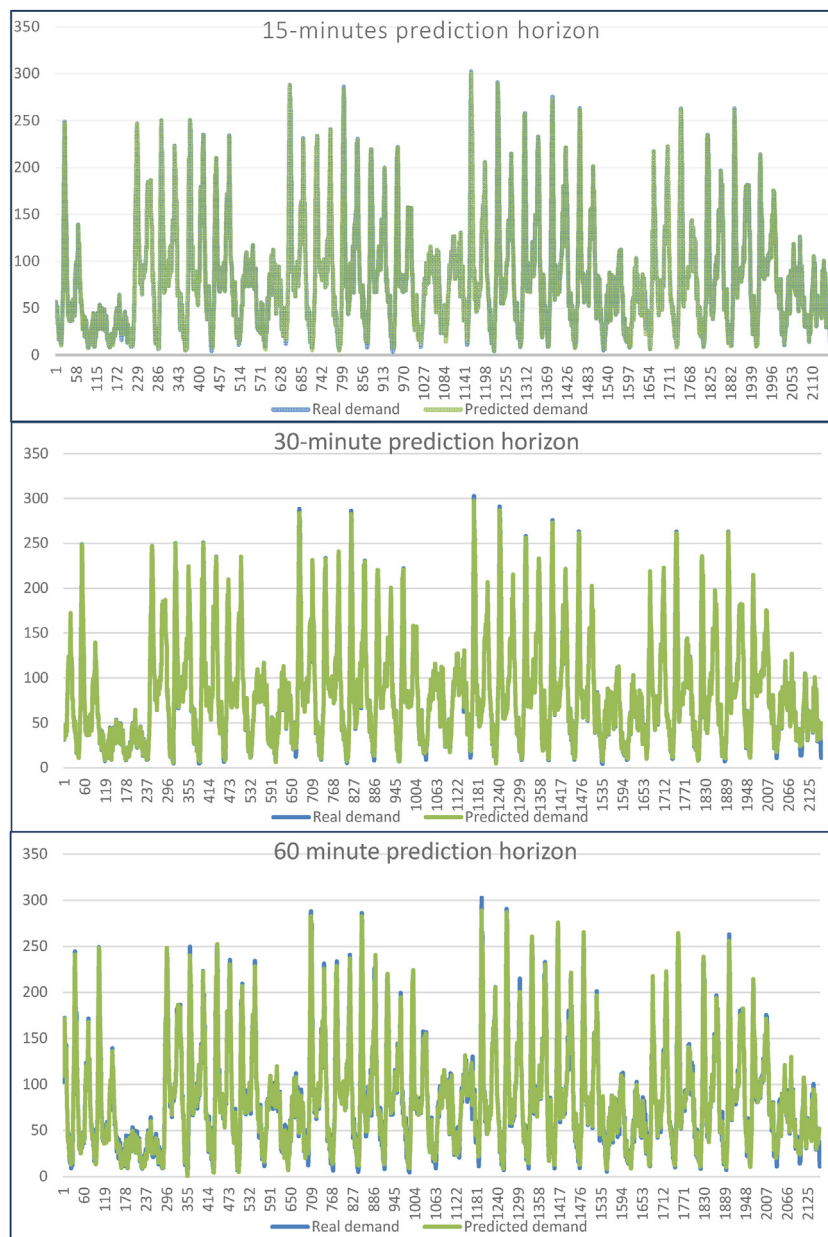
#### 4. Short term prediction accuracies

This section summarises the results of passenger demand prediction for bus routes.

##### 4.1. Model performance evaluation and validation

The results for demand prediction variations for bus routes are provided in Table 7. Here, green highlights represent the highest demand prediction accuracy, while yellow represents the second-best accuracy.

The BiLSTM models predicted bus demands with over 90% accuracy for 15-min and 30-min forecast horizons and approximately 80% accuracy for 1-h demand prediction. The results also show that the LSTM models, both uni-directional and bi-directional LSTM, have superior performance when compared to the other deep learning models such as RNN, GRNN, MNN, RBF, and DLBP models. For



**Fig. 7.** Graphical representation of actual to predict passenger demand for BiLSTM models for (a) 15-min, (b) 30-min, and (c) 60-min time horizons.

bus route predictions, the results showed that BiLSTM provided a forecasting MAPE accuracy of around 97–99% for a 15-min prediction horizon, 94.77–98.40% for a 30-min prediction horizon, and 87.61–94.38% for hourly prediction. Compared to LSTM, the BiLSTM methodology shows the highest prediction accuracy, which does not deteriorate substantially with longer forecast horizons. The accuracy of the highest prediction horizon, 60-min, remained high at 87.61–94.38% demonstrating the capability to capture the complexity for the hourly prediction horizon.

#### 4.2. Model fitting and prediction accuracy

Model fitting is measured with R-squared values. The BiLSTM model results showed a high correlation between actual and predicted demands as shown in Fig. 6. The LSTM model also showed over 0.9 R-squared values for 15, 30, and 60 min predictions. A comparison of the forecasting performance of the models is shown in Fig. 7 (a, b and c). These figures reflect the degree to which a model's prediction deviates from actual passenger demands on bus services. The hourly model has the lowest fitting degree, whereas the 15-min model shows the highest fitting degree.

**Table 8**

Comparisons of selected studies on bus passenger flow prediction.

| Authors                            | Method                       | Contrast method                                 | Method Style | Predict object | Data source   | Data                  |                          | Accuracy                                                              |
|------------------------------------|------------------------------|-------------------------------------------------|--------------|----------------|---------------|-----------------------|--------------------------|-----------------------------------------------------------------------|
|                                    |                              |                                                 |              |                |               | Temporal              | Spatial                  |                                                                       |
| Liyanage et al., 2021 (this study) | BiLSTM                       | DLBP, MNNs, RBFNs, GRNNs, RNNs                  | Single       | Line           | Smart-card    | 1-month               | 1,781 stops in 18- lines | MAPE 0.03–0.12<br>MAE 0.01–0.06<br>RMSE 0.03–0.08<br>Accuracy 92%–99% |
| Gong et al. (2014)                 | Kalman filter-based ARIMA    | Direct-addition                                 | Hybrid       | Stop           | APC and video | 2-months              | 3-stops in 1-line        | RE around 3%                                                          |
| Ma et al. (2014)                   | IMMPH with AR, SARIMA, ARIMA | ANNPH                                           | Hybrid       | Line           | AFC           | 1-year                | 1 line                   | MAPE 5.82%                                                            |
| Zhai et al. (2020)                 | HTSDBNE                      | ARIMA (2,1,2) ELM, TS-ANN, SLMBP, SAE-DNN, MPDF | Hybrid       | Stops          | APC           | 6-months              | 2-stations in 2-lines    | Error 8.787%                                                          |
| Xue et al. (2015)                  | IMM with ARMA, SARIMA, ARIMA | Real data                                       | Hybrid       | Line           | AFC           | 4-months              | 1-line                   | MAPE 9.084%                                                           |
| Pekel and Soner Kara (2017)        | POA-ANN<br>IWD-ANN           | GA-ANN                                          | Hybrid       | Line           | AFC           | NA                    | NA                       | MSE less 0.1                                                          |
| Zhang et al. (2017)                | GM(1,1)                      | Real data                                       | Single       | Line           | Manual survey | 6-consecutive Mondays | 1-line                   | RE less 10%                                                           |
| Bai et al. (2017)                  | MPDF                         | DBN, AE, AP-AE, FFNN, ARIMA                     | Hybrid       | Line           | Real data     | 5-months              | 1-line                   | MAPE 10.743%                                                          |
| Liu and Chen (2017)                | SAE-DNN                      | Real data                                       | Hybrid       | Stops          | AFC           | 4-months              | 4 stations in 7-lines    | MAPE over 75%                                                         |
| Zhou et al. (2013)                 | Poisson model ARIMA          | Real data                                       | Hybrid       | Stop           | APTS          | 5-months              | 416 bus stops            | Around 79%                                                            |
| Zhang et al. (2011)                | Kalman filter                | BP-ANN                                          | Single       | stop           | AFC video     | 1-month               | 4-stations               | Around 80%                                                            |
| Han et al. (2019)                  | LSTM<br>Nadam-SGD            | Naïve, ARIMA, SVR, 5 traditional LSTM           | Hybrid       | Stops          | SCD           | 1-month               | 30-stations              | MAPE 24.002%                                                          |

## 5. Comparative evaluation with other studies in the literature

This section presents a comparative evaluation between the findings from this study and several similar studies reported in the literature. While previous sections of the paper compared different models using the same data, this section presents a holistic view of the effectiveness of our models compared to other literature. The comparisons in Table 8 are for three categories of models based on “traditional classical algorithms”, “regression models”, and “machine learning-based models” for single and hybrid models. The table presents the method used in each study, the comparisons conducted against other models, the architecture (single versus hybrid) used, the prediction object (route/line or bus stop), data source and dimensionality (temporal or spatial), and the accuracy of the model as reported in each paper.

The accuracy results clearly show that nonlinear methods performed better than linear methods. The hybrid models, which combine different time series methods to capture challenging data patterns and characteristics in their datasets, are highly complex and generally do not perform as well as single models due to the weakening of the universality of the hybrid models as a result of their more complex architectures. Among more recent literature, non-linear approaches based on machine learning seem to have been widely adopted for bus passenger demand prediction.

Establishing multiple suitable indicators for measuring prediction accuracy is essential, as no single indicator can measure the full precision and/or bias of predicted values. The performance indicators used to compare different models were based on existing literature (Vandeput, 2019) and included three key performance measures: MAPE, MAE and RMSE. As per Table 8, the BiLSTM model proposed in this study shows high prediction capability compared to the previous literature with an MAPE error of 0.03–0.12, MAE error in the range of 0.01–0.06 and RMSE with 0.03–0.08 range. With this, the prediction accuracy is over 92–99% for three different time horizons. This study therefore contributes to the body of knowledge in this field by using BiLSTM models with evidence of much higher accuracies exceeding 90%.

## 6. Conclusions and directions for future research

This paper presented robust deep learning models that were developed for short-term temporal predictions of passenger demand

using real-world data. The data was obtained from the MyKi smart-card fare payment system in Melbourne. Deep learning models were constructed representing 15-min, 30-min, and 60-min. These models were developed using BiLSTM and LSTM deep learning methodologies based on one month of data comprising 27,823 data points for the 15-min model, 27,360 data points for the 30-min model, and 26,561 data points for the 60-min model. The findings of this study showed that both the BiLSTM and LSTM architectures provided the highest predictive intelligence accuracy of over 90% for short-term predictions of passenger demands for 15-min, 30-min and 60-min time horizons.

The main limitation of this study was limited access to smart-card data which in our case was constrained to one month of data between 1–27 May 2018. This was the only data made available to the researchers. Access to larger data sets covering more months and years as well as more routes can help improve the accuracy and reliability of the models even further. Furthermore, for this analysis, aggregated passenger demand for inbound and outbound direction for each route was necessary because the data did not have sufficient observations to enable separate analyses for inbound and outbound directions. It is noted here, however, that the direction of the service will not have an impact on model selection and performance. The key factor influencing performance, in either inbound or outbound directions, is the availability of quality representative data that can be used for model training and testing. We aim to address this limitation in future studies through analysis of large data sets that can provide accurate and sufficient observations in both directions of travel.

The AI-inspired deep learning passenger demand forecasting models can be applied in a wide variety of public transport operational contexts. In this work, we estimated future passenger demands based on routes. The same methodology can be applied to estimate passenger demands at bus and tram stops and also train stations where this information can be used to predict the number of buses, trams or even number of compartments on a train that are needed to meet passenger demands for different routes at different times of the day. Knowing this information in advance can save operators money through reducing costs of operations. It can also improve customer satisfaction and enhance the sustainability outcomes for urban areas through promoting reliable public transport modes of urban mobility and shifting drivers away from energy-intensive private vehicles to more energy-efficient modes of public transport.

Researchers interested in extending this work should also consider improving the performance and practicality of Bi-LSTM models for more data structures, such as daily, weekly, and monthly aggregations based on different aspects including route, service direction, and at points of boarding and disembarking such as bus and tram stops and at train stations. Future studies should also consider more patterns representing the impacts of particular weather conditions and traffic incidents. Other aspects of extending this research would be considering advanced deep learning architectures and the influence of network parameters on model performance.

## Ethics clearance

This work was carried out in accordance with Swinburne University's Human Research Ethics Guidelines. Project ref: 20214391–6022.

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